

Causal Scale Dynamics

Compact Reference to the Ruble Equations

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Status. Compact reference to the homogeneous theory. The scale-capacity equivalence law is a physical principle, not a theorem of mathematics. The covariant perturbation lift and global vacuum sector remain open.

The theory is most compactly stated as a sequence of definitions and one physical equivalence principle. The equations are organized by mathematical type rather than by discovery history.

Kinematic and geometric register

R1 — causal scale.

$$g_{\text{phys}} = \sigma^{-2}g, \quad N = -\ln \frac{\sigma}{\sigma_c} = \ln \frac{a}{a_c}.$$

The conformal metric g fixes causal cones; the scale section $\sigma \in \Gamma(\mathcal{E}[1])$ fixes physical calibration. N is logarithmic Weyl scale, not Newtonian absolute time.

R2 — scale tractor.

$$I_A = \frac{1}{4}D_A\sigma.$$

The scale tractor packages σ , its first derivative, and the trace combination of its second derivative.

R3 — tractor transport and norm.

$$(\nabla_a \nabla_b \sigma + P_{ab} \sigma)_0 = \frac{4\pi G}{c^4} \sigma T_{ab}^{\circ},$$

$$I^2 = \frac{2\pi G}{3c^4} T - \frac{\Lambda_g}{3}.$$

The first equation is local and trace-free. The second fixes the scalar calibration channel up to a global constant.

Horizontal state register

R4 — normal chirality.

$$Q = P_+ - P_-, \quad Q^2 = 1, \quad JQJ = -Q.$$

$P_\pm$ project onto the two null lines of the Lorentzian normal plane of a codimension-two cut. The homogeneous horizontal response is assumed to factor through this fundamental chirality quotient.

R5 — binary exponential family.

$$\omega_\theta = \frac{e^{\theta Q}}{2 \cosh \theta}, \quad \Psi(\theta) = \ln(2 \cosh \theta).$$

Its dual coordinates and metric are

$$\eta = \Psi'(\theta) = \tanh \theta, \quad G_{\theta\theta}^{\text{BKM}} = \Psi''(\theta) = \text{sech}^2 \theta.$$

The identity

$$\eta^2 + G_{\theta\theta}^{\text{BKM}} = 1$$

is the normalized binary moment relation $\langle Q^2 \rangle = 1$.

R6 — horizontal soldering.

Under the rank-one ratio and measurability hypotheses, the Connes cocycle chain rule gives

$$\theta = \varrho_\perp(N - N_c).$$

The theory selects the fundamental character

$$\varrho_\perp = 1.$$

This is a physical representation choice: the fundamental null-normal character is identified with the fundamental scale/inverse-scale character. It is not a theorem of Cauchy's equation, and it is not obtained from an integer restriction on conformal weights.

Scale-capacity closure

R7 — Ruble scale-capacity number.

Let $G_{NN}^\perp$ be the physical BKM norm of the selected horizontal scale tangent and S_c the horizon entropy of the self-dual wall. Define

$$\mathfrak{R}_c := \frac{k_B}{S_c} G_{NN}^\perp(N_c).$$

The Scale-Capacity Equivalence Principle is

$$\boxed{\mathfrak{R}_c = 1.}$$

In canonical binary coordinates this is equivalently

$$\boxed{\frac{k_B}{S_c} G_{NN}^\perp(N) = \text{sech}^2(N - N_c).}$$

It states that the fundamental Weyl translation is isometric, at self-duality, to the fundamental horizontal state translation when the BKM metric is normalized by horizon entropy.

R8 — modular free-energy source.

$$\boxed{\rho_X(N) = \frac{k_B T_c}{2V_c} G_{NN}^\perp(N).}$$

This is the constitutive law. It converts dimensionless relative-entropy curvature into energy density by the physical modular temperature and causal-wall volume. The factor 1/2 is the Taylor coefficient of relative entropy at coincidence.

R9 — Hawking–Friedmann conversion.

For a flat four-dimensional FLRW apparent horizon,

$$\boxed{R_c = \frac{c}{H_c}, \quad \frac{S_c}{k_B} = \frac{\pi R_c^2 c^3}{G\hbar}, \quad k_B T_c = \frac{\hbar c}{2\pi R_c},}$$

and

$$\boxed{k_B T_c \frac{S_c}{k_B} = E_{\text{MS},c} = \rho_{\text{crit},c} V_c.}$$

R10 — closed source.

Combining R7–R9 gives

$$\boxed{\rho_X(N) = \frac{1}{2} \rho_{\text{crit},c} \text{sech}^2(N - N_c).}$$

At the crossing,

$$\boxed{\Omega_{X,c} = \frac{1}{2}, \quad \rho_X(N_c) = \rho_{\text{ordinary}}(N_c).}$$

The exact equality is with the complete non-dark sector. Relative to dust alone,

$$\frac{\rho_X(N_c)}{\rho_m(N_c)} = 1 + \frac{\rho_r(N_c)}{\rho_m(N_c)} + \dots$$

Dynamical consequences

R11 — equation of state.

Separate conservation,

$$\rho'_X = -3(1 + w_X)\rho_X,$$

gives

$$w_X(N) = -1 + \frac{2}{3} \tanh(N - N_c).$$

R12 — shape invariant.

$$9(1 + w_X)^2 + 6w'_X = 4.$$

Equivalent forms are

$$\frac{\rho_X}{\rho_*} + \frac{1}{4} \left(\frac{d \ln \rho_X}{dN} \right)^2 = 1,$$

and

$$X' = \frac{2}{3} - \frac{3}{2}X^2, \quad X := 1 + w_X.$$

R13 — self-dual comparison.

The symmetrized relative entropy of the state and its modular reflection is

$$\mathfrak{S}_J(\theta) = 4\theta \tanh \theta.$$

It has a unique global minimum at $\theta = 0$. Along the rigid history,

$$\mathfrak{S}_J = 6(N - N_c)(1 + w_X).$$

Thus relative-entropy positivity fixes the orientation of the $w_X = -1$ crossing relative to increasing Weyl scale.

R14 — central blindness.

$$\text{Var}(K + \alpha \mathbf{1}) = \text{Var}(K).$$

The local response is blind to the absolute energy zero. A global scalar lift remains in Λ_g .

What is law, what is solution data, and what is measured

Dependency structure of Causal Scale Dynamics

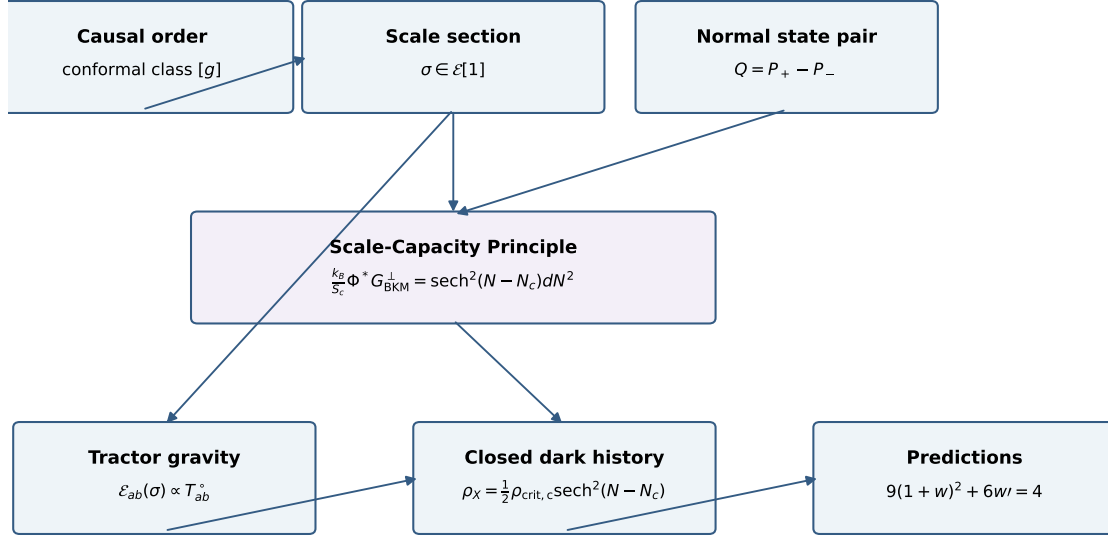


Figure 1: Dependency structure of the theory. The scale-capacity principle is the single new equivalence law closing the homogeneous amplitude.

Quantity	Type	Role
$[g]$	causal/conformal geometry	fixes null cones and causal order
σ	scale section	metric calibration
N	Weyl coordinate	logarithmic scale displacement
Q	normalized normal chirality	fundamental horizontal binary score
θ	horizontal state coordinate	relative modular polarization
$\mathfrak{R}_c$	dimensionless equivalence number	entropy-normalized peak BKM speed; postulated value 1
N_c	solution origin	location of the intrinsic self-dual event relative to today
Ω_{m0}, Ω_{r0}	measured state data	determine the location of the crossing in the closed background
Λ_g	global lift/sector	not part of local trace-free transport; zero in the open-future branch

The theory therefore contains no continuously fitted dark-history function and no independent dark amplitude. It does contain one explicit physical equivalence principle and one global sector choice.